

S.No	Title of the Paper	Author(s)	Journal / Conference	Problem Addressed	Methodology	Strengths	Limitations	Research Gap
1	Hierarchical Graph Network for Multi-hop Question Answering	Yuwei Fang, Siqi Sun, Zhe Gan, Rohit Pillai, Shuhang Wang, Jingjing Liu	Proceedings of EMNLP 2020	Multi-hop question answering over multiple Wikipedia documents requiring reasoning across paragraphs, sentences and entities.	Constructs a hierarchical graph with Question-Paragraph-Sentence-Entity nodes initialized using RoBERTa + BiLSTM and performs message passing using a Graph Attention Network (GAT) . Jointly predicts supporting paragraphs, supporting facts, entities and answer spans.	Explicit graph-based reasoning, strong supporting-fact prediction, state-of-the-art performance on HotpotQA, interpretable graph reasoning.	Limited to text-only Wikipedia documents, extractive QA, relies on hyperlinks and structured text, no spoken audio or explainability beyond attention weights.	LISTEN extends HGN to spoken audio , replaces text passages with timestamped transcript segments, uses a segment-level evidence graph, integrates SHAP explanations and introduces counterfactual node-removal validation. (ACL Anthology)
2	Explainable Multi-Hop Question Answering: A Rationale-Based Approach	Kyubeen Han, Youngjin Jang, Harksoo Kim	Big Data and Cognitive Computing (MDPI), 2025	Improve explainability in multi-hop QA without requiring manually annotated rationale labels.	Uses ELECTRA encoder with a Pointer Network to iteratively select supporting rationale sentences while enforcing semantic consistency between selected evidence and predicted answers.	Generates faithful explanations without rationale annotations, lightweight architecture, strong rationale selection performance.	Works only on textual documents, no speech processing, no graph neural reasoning, no interactive explanation.	LISTEN applies explainability to spoken audio , combines GAT reasoning with SHAP explanations and interactive counterfactual validation over timestamped evidence. (MDPI)
3	Thoughtful and Cautious Reasoning: A Fine-Tuned Knowledge Graph-Based Multi-Hop Question Answering Framework	Yinghao Zheng, Ling Lu, Yang Hu, Yinong Chen, Aijuan Wang	Engineering Applications of Artificial Intelligence (Elsevier), 2025	Multi-hop reasoning over knowledge graphs while reducing semantic ambiguity and noisy reasoning.	Proposes TCR-KGQA using a novel question encoder, knowledge graph embeddings and a residual-GRU instruction update framework for iterative reasoning.	Rich semantic understanding, effective noise suppression and accurate graph reasoning.	Requires structured knowledge graphs, cannot process spoken audio or automatically construct evidence graphs from transcripts.	LISTEN automatically builds evidence graphs from Whisper/Wav2Vec2 transcripts and performs GAT-based reasoning with explainable graph visualization and counterfactual validation. (ScienceDirect)

4	Spoken Question Answering and Speech Continuation Using Spectrogram-Powered LLM	Eliya Nachmani, Alon Levkovitch, Roy Hirsch, Julian Salazar, Chulayuth Asawaroengc	ICLR 2024	End-to-end spoken question answering directly from speech without relying entirely on text transcripts.	Uses a speech encoder operating on spectrograms connected directly to an LLM for spoken QA and speech continuation.	End-to-end speech understanding without explicit ASR dependency, strong spoken QA capability.	Focuses on single-hop spoken QA, lacks retrieval over long recordings, graph reasoning, evidence attribution and explainability.	LISTEN extends spoken QA by introducing retrieval across long audio archives, graph-based multi-hop reasoning, evidence graphs and counterfactual explanations.
5	Question Calibration and Multi-Hop Modeling for Temporal Question Answering	Chao Xue, Di Liang, Pengfei Wang, Jing Zhang	AAAI Conference on Artificial Intelligence (AAAI 2024), pp. 19332–19340	Prior temporal KGQA models focus on entity information but ignore temporal entity transfer, and fail to explicitly model multi-hop graph relationships	Calibrates question representation by fusing it with time-constrained KG concepts, then applies a GNN layer for multi-hop message passing before generating the answer	Addresses two concrete gaps (temporal blindness + multi-hop reasoning) in one framework; strong benchmark venue (AAAI)	Tested only on standard benchmarks, unclear scalability to large/noisy KGs; reviewers have flagged near-identical pipelines/results with Liu et al. (2023) and Zha et al. (2024), raising nove	Little work on applying calibration + multi-hop reasoning to noisy or streaming real-world temporal KGs